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Review Article

A review and appraisal of clinically prevalent control devices for electrically powered wheelchairs

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Abstract.
BACKGROUND: Electrically powered wheelchairs (EPW) enable individuals who are not capable of walking or propelling a manual wheelchair to have independent mobility. Control devices enable users to direct the EPW to move at the desired speed and direction.
OBJECTIVE: To review and appraise the current use of EPW control devices to assist clinicians when selecting the most appropriate device for the needs of individual users.
METHODS: Two databases were searched returning 273 unique articles for review. The prescription forms of the 20 EPWs available on the Scottish NHS contract were reviewed to determine availability.
RESULTS: Based on four studies, the most prevalent devices were hand joystick, sip and puff and chin joystick (84%, 8% and 8% respectively). However, terminology was inconsistent, hampering comparison, and population groups non-representative. The appraisal of prescriptions forms found hand joysticks to be the standard option. Sip and puff and chin joysticks were available on 15% and 35% of EPW models respectively.
CONCLUSIONS: The standard, hand joystick (position sensing and proportional) meets the control device needs of the vast majority of current EPW users. Further studies are required to quantify the needs of those unable to use the control devices that are currently available.

Keywords: Electrical powered wheelchair, control device, powered mobility, hand joystick, chin joystick, sip and puff control

1. Introduction

An electrically powered wheelchair (EPW) enables individuals who are not capable of propelling a manual wheelchair to have independent mobility. The EPW occupant uses a control device to direct the wheelchair to move at the desired speed and in the desired direction of travel. It is essential that the control device is easy to use to facilitate efficient mobility. If the control device is inadequately matched to the user then the EPW is unlikely to be accepted, therefore greatly reducing that individual's freedom.

EPWs were first developed at the start of the 20$^{\text{th}}$ century [1]. The demand for a wheelchair that could be driven without significant manual force was increased by three main factors; developments in healthcare for treating individuals with long term conditions, the expansion of rehabilitation and the growth of a disability movement [2]. One of the first patents for an EPW was approved in the USA in 1940 and featured an electric switch control device [3]. The original control devices fitted to EPWs were hand joysticks that were connected to a system of four switches [2]. These systems were unreliable and made manoeuvring difficult [1].

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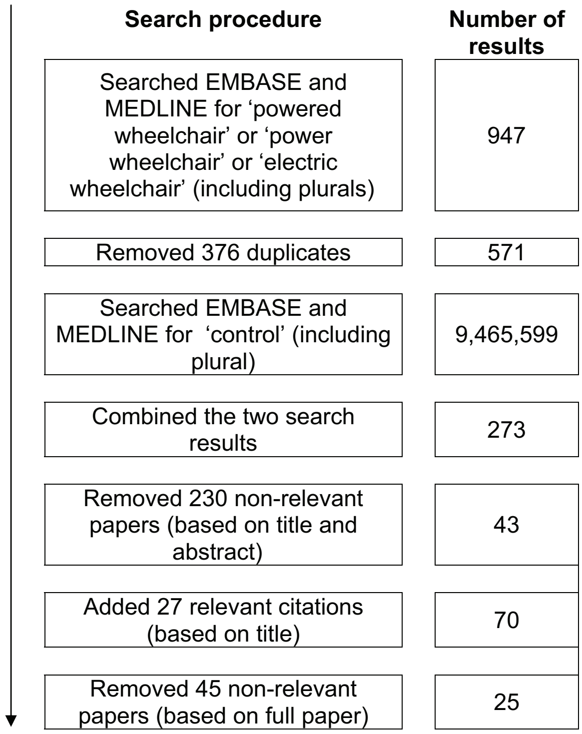

Fig. 1. Flow diagram of article search.

To overcome the difficulties with a switch system, the aerospace company Penny and Giles (P&G) developed a proportional hand joystick control device specifically for EPWs in the late 1970s [2]. This and subsequent, similar controllers utilised transistors and these provided better control and allowed for the development of specialised control devices for the user [1]. In 1988, the New Zealand company, Dynamic Controls, introduced its first microprocessor into its controller design. This enabled increased reliability, improved safety, a decrease in the number of components and a reduction in production costs [2]. More recently, as microprocessors have decreased in size, become increasingly powerful and less expensive, joysticks have become more reliable, more sensitive and smaller.

1.1. *Purpose of the study*

The purpose of this study was to review and appraise the current use of control devices for EPW to assist clinicians when selecting the most appropriate device for the needs of individual users. The aims were to:

- Identify the prevailing control devices in routine use clinically;
- Profile the identified control devices;
- Compare and contrast these control devices.

2. Methods

A literature search was conducted using the Medline (1946 to July 2013) and EMBASE (1974 to July 2013) electronic databases. English language articles containing the terms 'electric wheelchair' or 'powered wheelchair' or 'power wheelchair' and the term 'control' were identified. These terms were chosen to cover all the commonly used terms for EPWs [4]. All duplicates were removed and the abstracts were scrutinised by the authors to check that they contained relevant information about EPW control devices. If the title and abstract provided inconclusive information, the full-text version was retrieved to decide on inclusion. Only peer reviewed journal publications and books were selected to ensure that information was complete and reliable. The reference lists of all relevant articles were also checked. The required information was extracted from the identified articles for analysis.

All the current prescription forms for EPWs available on the National Health Service (NHS) Scotland-wide procurement contract were reviewed. This was undertaken to determine the range and types of control devices that were available to the five NHS wheelchair and seating services in Scotland [5]. The contract was established following a formal, open tendering process undertaken by NHS National Procurement [6] consistent with the Scottish Government's public sector procurement requirements that meet the legal requirements of the European Community [7]. The wheelchair specification requirements were drawn up by a panel of recognised experts in the field and the subsequent tender evaluation was undertaken by the same panel. The contract covers indoor-only, indoor-outdoor and outdoor-indoor EPWs. It does not cover outdoor-only EPWs.

The terminology employed was based on the international wheelchair standards, ISO 7176-26 Part 26 – Vocabulary [8] and ISO-7176-14 Part 14 – Power and control systems for electrically powered wheelchairs and scooters [9]. The former, although not widely adopted and not fully comprehensive [4], offers the only international recognised vocabulary for describing wheelchair components, including drive systems, and postural support devices. The latter, does not formally define any additional vocabulary that has been used in this article, but does employ terminology used in this article that has not been formally defined by either standard. In the context of this article, the term 'user' is used to denote the person who is both supported by the wheelchair seating system and operates the EPW [4].

Table 1
Definitions of EPW control devices

Control device		Definition
Hand joystick	Proportional	A position sensing control device where the magnitude of the control output is proportional to the amount of deflection of a lever (knob) from the central position.
	Switched	A position sensing control device with typically four different discrete control positions for forward, rear, left and right directions.
	Isometric	A force sensing control device where no range of motion is required to operate, instead the magnitude of the control output is typically proportional to the force applied.
	Miniature	As a proportional hand joystick, except that the lever (knob) requires a much smaller force to deflect it from the central position and the amount of deflection required is also reduced.
Finger joystick		A proportional control device that uses infrared light to sense finger movement.
Sip and puff		A control device that detects the sucking or blowing of air down a straw using pneumatic switches to detect pressure differentials.
Chin joystick		A position sensing control device mounted so that the user can control it with their chin. The amount of deflection from the central position is proportional to the magnitude of the control output.
Eye gaze		A control device that monitors the user's eye movements and these can be translated into a control output.
Tongue pad		A control device located in the user's mouth that can be operated using the tongue.
Head/hand/foot switch	Single switch	A single switch that can be operated using the head/hand/foot. Single switch control has to be done via scanning using a specialised control unit.
	Switch array	An array of switches that can be operated using the head/hand/foot. Typically 5 different switches enabling power on/off, forwards, backwards, left and right directions.

3. Results

The literature search (up to 29$^{\text{th}}$ July 2013) yielded 273 citations after duplicates were removed (Fig. 1). Scrutiny of the titles and abstracts of these citations for relevancy removed 230 articles, leaving 43 articles that were retrieved for further screening. Examination of the reference lists of these added 27 articles, for a total of 70 potentially relevant articles. The identified articles were further reviewed for relevance. This led to the exclusion of 45 articles leaving 25 relevant articles. Reasons for exclusion included the use of non-common, novel control devices that were still only at the research and development stage.

The review of the EPWs available to the NHS in Scotland found that there were four different manufacturers, with a total of 20 models, and with control devices made by five different manufacturers. Control device definitions were created to enable easy comparison between control devices. These were based, as far as possible, on existing ISO terminology and are presented in Table 1.

3.1. Control devices in routine use

Four studies were found that reported on the prevalence of control devices for EPWs. The earliest, Lachmann and co-workers [10], reported an exercise to establish the level of need for wheelchair seating by proactively seeking referrals for assessment. The 215 referrals judged to meet the selection criteria represented around 6% of the total wheelchair service users in the area covered (Cambridge, UK). Forty one percent were male, 40% had cerebral palsy (CP), 16% had multiple sclerosis (MS) and 9% had been subjected to a cerebrovascular accident (CVA). Sixty two (29%) were under 18 years old, 80 (37%) were 18 to 64 years old and 73 (34%) with 65 years or older. Sixty four people were EPW users. Unfortunately, the demographic details were not broken down by type of wheelchair. The vast majority of EPW users, 60 (94%) used 'standard controls' that would, presumably, be a proportional hand joystick control. The reminder, 2 (3%) and 2 (3%) used control devices activated by the chin and arm or tongue respectively. It was reported that overall 40 (63%) rated their controls 'as good as can be' while 16 (25%) rated their control system as 'could be better'. Ten users (16%) could no longer use their EPW or were described as having great difficulty.

A survey by Berkowitz and co-workers [11], primarily concerned with costs, questioned 500 people with spinal cord injuries in the USA. Eighty three percent were male and 51% were tetraplegic. Twenty-nine percent of the people surveyed used an EPW as their primary wheelchair. Of these, 88% used a 'joystick', 5% used 'sip-and-puff', 2% used 'chin control', 1% used 'head control' and 4% were unknown. For the purposes

Table 2
Percentage of users using each different type of control device

	Population	Control device used (%)						Comment
		Hand control joystick	Sip and puff	Chin control	Head control	Chin or head control**	Other*	
Lachmann et al., 1993 [10]	64	94	0	3	0	3	3	Restricted to those referred for a seating assessment. Population details not broken down for EPW users.
Berkowitz et al., 1998 [11]	140 with SCI	88	5	2	1	4	0	Restricted to SCI and those using an EPW as their primary wheelchair. One of hand joysticks known to be used with T-bar attachment. Five unknowns not included.
Fehr et al., 2000 [12]	30 institutions	81	9	–	–	9	1	Reported results weighted according to the size of institutions. Total population number not reported. Data for head and chin control reported as one.
Dolan and Henderson, 2012 [13]	27 (13 with MS, 6 with CP, 8 other)	100	0	0	0	0	0	Restricted to only those attending a seating clinic for adults for postural support in their wheelchair.
Simple average	N/A	92	4	2	0	4	1	Average of the four studies.
Weighted average	1731	84	8	2	1	8	1	Weighted according to population numbers, assuming average institution size of 50 for Fehr et al., 2000 [12].

*Other includes: ‘arm’, ‘tongue’, ‘tongue pad’, ‘head, hand and foot switch’ and ‘eye gaze’; **Percentages for chin and head control were combined for those reported separately to allow comparison with single percentage reported by Fehr et al. 2000 [12].

of this study, ‘joystick’ was assumed to be a hand joystick.

Fehr and co-workers (2000) surveyed 200 healthcare institutions, including comprehensive rehabilitation hospitals, veteran centres, blind centres, geriatric centres, wheelchair vendors, that had facilities for patients with multiple sclerosis, amyotrophic lateral sclerosis, muscular dystrophy, brain injury, spinal cord injury, development disability or cerebral palsy. No details on the actual breakdown of the patient sample population were presented, nor on the overall number of patients covered. Thirty of the returned questionnaires were complete enough to be included in the reported results on control devices usage. The respondents were asked to report on the percentage of the institution’s regular users of EPW using five difficult control devices (‘joystick’, ‘chin control’, ‘eye gaze control’, ‘sip and puff’, ‘head control’) with an option for other. A high majority of EPW users, 81% used a ‘joystick’, 9% used a head or chin control, 9% used sip and puff and 1% used other devices. It was also reported that 85% of respondents said that they evaluated patients for whom EPWs were not an option as they could not control them due to their lack of motor skills, strength or visual acuity. It was estimated that 18–26% of non-ambulant wheelchair users who could not self-propel a manual wheelchair were also unable to operate an EPW.

In the most recent study, Dolan and Henderson [13] reported that all 27 (48% with MS, 22% with CP, 30% with other primary diagnoses) of those using an EPW who attended at a seating assessment clinic in Edinburgh, Scotland, used a ‘proportional, position sensing, hand joystick (side mounted)’. All the attendees were adults (18 years or over).

The results of the four studies were complied and are presented in Table 2. The simple average is the average of each of the four studies. To take account of the differences in population sizes of the four studies a weighted average was also calculated. As Fehr and co-workers [12] did not report the number of EPW users per institution, it was necessary to use an estimate of 50 users per institution, thus giving an overall population of 1731. The vast majority of EPW use a hand joystick control, in the region of 84 to 92% (weighted and simple average respectively). The second most prevalent was sip and puff control which accounted for between 4 to 8%. Chin or head control also accounted for between 4 to 8%, though for the studies that reported separate data, chin control was more prevalent at around 2% compared to 0 to 1%. All other control devices account for only around 1%.

Table 3
Availability of control devices on the NHS Scottish contract

		Manufacturer				
		A	B	C	D	All
No. of models		1	1	11	7	20
No. of control device manufacturers used		2	2	3	4	5
No. of models with specified control options	Proportional hand joystick (sidearm mounted)*	1	1	11	7	20
	Attendant control**	1	1	9	7	18
	Dual control	0	1	10	7	18
	Proportional hand joystick (centre mounted)	0	1	5	5	11
	Chin joystick	0	0	4	3	7
	Proportional hand joystick (tray mounted)	1	0	5	0	6
	Isometric joystick	0	0	0	5	5
	Sip and puff	0	0	0	3	3
	Switch array	0	0	0	3	3
	Finger control	0	0	0	2	2

*Available at inclusive cost on all models; **Available at inclusive cost on 4 models only.

3.2. Availability

The review of the EPWs available to the NHS in Scotland revealed that there were four different manufacturers of EPW with a total of 20 EPW models (Table 3). Control devices from five different manufacturers were available, though not all options were available on all models. For all the models, the inclusive cost option was a proportional hand joystick mounted on side of wheelchair at front of the arm support. Seventeen (85%) of models had an option of attendant hand joystick control, though for all but 4 models this was at an additional cost, either due to the requirement for a different control device and/or the different mounting requirements (on the rear of the wheelchair adjacent to one of the push handles if available). Eighteen (80%) of models had an option of dual control (with both occupant and attendant hand joysticks) all at an additional cost (due to the need to duplicate the control devices). Centre mounted and tray mounted hand joystick controls were available on 11 (55%) and 6 (30%) models respectively. All were at additional cost, presumably due to the additional bracketry and tray required, but also due to the need for a separate joystick with low depth for clearance. All other options, sip and puff, isometric joystick, switch control and finger (steer) control were only available as additional cost options from one manufacturer, though it would be possible to add these control devices to some of the other manufacturers' models retrospectively.

3.3. Prevalent control devices

The three most prevalent control devices, hand joysticks, sip and puff control and chin joysticks, are described and reviewed below. The results of the review of the literature are summarised in Table 4.

3.3.1. Hand joysticks

Hand joysticks are by far the most commonly used control device EPWs. Traditionally the main types have been proportional, switched and isometric [1] but more recently the miniature joystick (Table 1) that requires only a reduce amount of deflection has become available. The proportional joystick is found on most EPWs and supplied as standard by manufacturers. Although three of the four studies summarised in Table 2 do not distinguish between the four types, based on availability and the authors' experience, it would be expected that nearly 100% of those in use are position sensing and proportional.

Hand joysticks are usually mounted on the side of a wheelchair at the front of the arm support with fore-aft adjustment. They can also be mounted centrally and/or incorporated in or on a lap tray, though this is only possible if the underside of the device does come into contact with the user.

Proportional joysticks work on the principle that the output sent to the EPW is proportional to the amount of deflection of the joystick from the centre position [1]. The joysticks use either potentiometers, variable inductors or optical sensors to convert displacement to voltage [14]. The amount of deflection required to achieve a full output signal can be varied by programming the EPW's microcontroller. A dead zone is essential whereby a small amount of joystick deflection results in zero movement of the EPW. The efficient use of a proportional joystick requires a certain amount of intact proprioception, joint mobility, dexterity [15], and the ability to maintain a constant grasp [16]. If a person has excessive movements or tremors in the hand, limited range of motion, athetoid movements, or spasticity rigidity these factors can reduce or prohibit control of an EPW [17].

Table 4
Summary of literature review findings for hand joystick, sip and puff and chin joystick

Control device	Article	Reported findings or observations
Hand joystick (conventional and isometric)	Stewart et al., 1992 [16]	Participants with cerebral palsy with prior experience of using a conventional hand joystick do not appear to benefit by the use of an isometric joystick.
	Cooper et al., 2000 [17]	(For conventional hand joysticks) excessive intention tremor, limited range of motion, athetoid motions, and spastic rigidity can reduce or prohibit control. The isometric joystick was demonstrated to be superior to the conventional hand joystick for driving in straight line and driving in a circle.
	Cooper et al., 2000 [18]	Despite the lack of experience with isometric joysticks, participants were found to perform better with the isometric joystick during manoeuvres.
	Cooper et al., 2002 [19]	With an isometric joystick, motion triggered reflexes may be less likely to occur.
	Dicianno et al., 2006 [20]	Anecdotally, most participants claimed that they preferred using a conventional hand joystick in comparison to an isometric joystick because it was a familiar device.
	Dicianno et al., 2009 [21]	People with certain disabilities have difficulty using conventional hand joysticks because of varying levels of tremor. Customised isometric joysticks may be superior to conventional hand joystick control for individuals with tremor.
	Dicianno, 2010 [15]	Efficient use of a conventional hand joystick requires a certain amount of intact proprioception, joint mobility, and dexterity.
	Mahajan et al., 2011 [22]	Participants were able to complete tasks faster with a conventional hand joystick in comparison to an isometric joystick. During forward driving fewer driving errors were recorded for the isometric in contrast during reverse driving fewer were recorded for the conventional hand joystick.
	Dicianno et al., 2012 [23]	The isometric joystick was preferred over the conventional hand joystick in the study.
Sip and puff	Jones et al., 2008 [27]	Can be prone to false activation due to shocks and vibrations.
	Winyard, 1976 [30]	Tetraplegia often necessitates suck/blow controls which are inherently more awkward to manage.
Chin joystick	Lipskin, 1970 [28]	Utilizes natural body motion that correlates with corresponding wheelchair functions. Quite reliable, functioning without maintenance for reasonable periods. Even simple commercial devices require substantial modification or customisation and access to engineering facilities. Chin cup requires custom contouring.
	Seamone and Schemeisser, 1985 [29]	Chosen for its positive control and good resolution capability. Overall found to be practical but concerns raised about being obstructive. Some users found that required too much neck/head movement. Incompatible with reclining back support and with position of user varying (due to fatigue or otherwise).
	Winyard, 1976 [30]	Tetraplegia often necessitates chin joystick controls which are inherently more awkward to manage.
	Murphy, 1978 [31]	Its size and location in front of the face are undesirable features of this system.
	McClay, 1983 [32]	Once patients at this rehabilitation centre progress from a chin-control box in front of their faces to a hand-control unit on the armrest, they have reported improvement in self-image, as they believe they appear (and truly are) less disabled and dependent. Also, most patients rely on leaning on the control box for balance. Once the box has been removed, they develop better sitting balance which consequently increases their confidence in the chair.
	Tew, 1988 [33]	Its position often results in a poor appearance and causes considerable inconvenience.

If an individual cannot use a proportional joystick then a switched joystick is an alternative option. A switched joystick is discretely controlled and usually has four switches, which using combinations of either single or double switch activations can create eight different discrete states [1]. However, these are inherently a less subtle and slower means of controlling an EPW.

Isometric joysticks have been developed to overcome the problems that can occur with proportional controllers for individuals with movement disorders [15,16]. Isometric joysticks sense the force being exerted on them and consequently, there is no requirement for a large displacement of the joystick. There have been numerous studies comparing the performance of the isometric joystick with the traditional position sensing joystick [15–23]. All studies have been laboratory based, using simulators or test tracks and able bodied and/or experience users with various medical conditions. Positive results were reported for the isometric joystick and it consistently matched or outperformed the performance of the conventional hand joystick in all except one [16] of the above studies. In contrast to the conventional hand joystick the isometric joystick was demonstrated to potentially minimize the effects of tremor in joystick control [21]. Anecdotally it has been reported that participants preferred the conventional hand joystick because it was a more familiar device [20]. Despite the high number of studies being conducted using isometric joysticks, they remain elusive with very limited commercial availability. This may be because of the availability of miniature joysticks that require a much smaller force to deflect but are still movement sensing joysticks.

A miniature joystick was used in a study by Pellegrini and co-workers [24], which demonstrated that replacing a traditional proportional hand joystick with an alternative control device (either a mini-joystick, an iso mini-joystick, a finger joystick or a pad that uses touch sensitive technology to determine finger position) enabled patients with muscular dystrophy to regain EPW driving ability. The study also demonstrated that a restricted ability to drive an EPW is significantly correlated with a decrease in key pinch strength and weakly correlated with an advance in age.

Even with the development of new, easier to use joysticks and associated controller systems, hand control joysticks are not suitable for all individuals; including some patients with advanced MS or motor neurone disease (MND), severe CP and high-level spinal cord injury (SCI). These users can potentially lack the strength, dexterity and control in their hands required to operate the joysticks.

3.3.2. *Sip and puff control*

Sip and puff control works by the controlled pushing and pulling of air through a straw. These actions can be further differentiated by how strongly or softly the air is pushed in or pulled out [25]. Usually the control method involves four or five different actions that can enable the control of an EPW. To enable normal breathing once the user has achieved the desired control, he or she ceases blowing or sucking and the selected control option is maintained [26]. This means that when a control signal is initiated the subsequent action continues to occur until an addition control signal is received to either cancel the action or change the type of action. This is a disadvantage because as a consequence the EPW motors could be instructed by a sip and puff controller to propel the wheelchair forwards and if the user is inactive after this signal has been initiated the EPW will continue to move forwards until a collision occurs, unless there is a stop switch and/or a time out. Another disadvantage of the sip and puff switch is that they can be prone to false activation due to shocks and vibrations [27], which could occur when travelling over uneven ground.

3.3.3. *Chin joystick*

Chin joysticks are essentially the same as proportional hand joysticks though they are generally smaller and more sensitive. The lever or knob is often a round disc with shallow concave. The joystick is mounted under a user's chin so that the user can operate the EPW with natural head and neck movements [28].

A chin joystick needs to be mounted in the correct position for the user and this usually consists of a rigid, metal, swing away mount that attaches to the back of the EPW. The optimum position for a chin joystick is determined by the user with the help of the clinician or clinical technologist setting up the system. The chin joystick is generally a reliable piece of equipment and can function without maintenance for reasonable periods of time, however, access to engineering facilities is generally required to modify or customise a chin joystick [28]. If the chin joystick at any point is not repositioned correctly into that optimum position then the user may struggle to control their EPW. There is also a requirement to operate additional switches for on/off and mode/speed profile change.

There are multiple potential reasons why patients cannot use a chin joystick; these include a lack of control and range of motion of the head and neck muscles [29]. A patients may also choose not to use a chin joystick because it is awkward to use and require an object to be placed in close proximity to their face [29–33].

3.4. *Comparison of control devices*

The advantages and disadvantages and indications and contra-indications are summarised in Table 5 and these are based on the literature review and the authors' own experience of using these control devices.

Eight factors were identified pertaining to the utility of each of the three most commonly used control devices. For comparison, these are presented in Table 6.

Table 5
Summary of control device advantages, disadvantages, indications and contraindications

Control device		Advantages	Disadvantages	Indications	Contraindications
Hand joystick	Proportional	– Inexpensive – Efficient – Intuitive – Readily available	– Position can limit access to tables – Not suitable for users who cannot maintain a constant grasp [16]	– Good hand muscle strength – Good fine motor control of hand – Good proprioception, joint mobility and dexterity [15]	– Limited range of motion [17] – Poor hand muscle strength – Poor fine motor control of hands – Poor proprioception, joint mobility and dexterity [15] – Hand tremors [17,21] – Spasticity in upper limbs [17] – Increased resting muscle tone, hyperexcitable reflexes, dystonia and clonus [23]
	Switched	– Inexpensive	– Time consuming	– Poor fine motor control of hand	– Very weak hand muscles – Spasticity in upper limbs – Hand tremors
	Isometric	– Efficient – Less fatiguing	– Not widely available [15] – Expensive	– Poor fine motor control of hand – Upper limb spasticity [19, 23] – A range of movement disorders e.g. Arthritis and arthrogryposis [16]	– Very weak hand muscles
	Miniature	– Efficient – Intuitive – Readily available – Less fatiguing	– Expensive	– Poor fine motor control of hand	– Very weak hand muscles – Poor proprioception – Hand tremors – Spasticity in upper limbs
Sip and puff		– No use of hand or neck muscles required	– Time consuming – Latched control can be potentially dangerous – Prone to false activation when travelling over uneven terrain [27] – Awkward to use [30]	– Poor fine motor control of hand – Spasticity – Hand tremors – Small neck range of motion – Weak neck muscles	– Poor breathing control – Tracheotomy
Chin joystick		– Efficient – No use of hand muscles required – Reliable [26] – Natural movements [26]	– Obtrusive [31] – Chin cup must be custom contoured [26] – Tiring – Less intuitive – Awkward to use [30] – Hinders sitting balance [32] – Poor appearance/ obstructive [29,33]	– Paralysed or very weak hand muscles – Good neck and head range of motion [29]	– Limited neck and head range of motion [29] – Weak neck muscles – Position changing due to fatigue [29]

A 5-point scale of 1 to 5 was used for comparison purposes with 1 being worse and 5 being better. The scores were determined by the authors based on the results presented in Tables 4 and 5. A comments column also explains the rationale behind the scoring for each factor. In summary, the hand joystick scores the highest for every factor. The chin joystick scores slightly higher than the sip and puff control, 24 and 20 respectively, and in particular it was rated as safer, more intuitive to use, more efficient and less expensive.

4. Discussion

The analysis of prevalence indicated that between 84 and 92% of users have hand control joysticks. How-

Table 6
Comparison of control devices

Factors	Control device			Comments
	Hand joystick	Sip and puff	Chin joystick	
Safety	4	2	3	Standard location of hand joystick on side of wheelchair prone to unintended activation and impacts. Sip and puff works on a latch control basis and therefore is not as safe as a control system with an automatic return to neutral feature. Sip and puff and chin control bracketry close to head and neck of user.
Intuitive	5	1	3	Hand joysticks are familiar devices due to their popularity in video game controllers. Training required for sip and puff. Chin joystick often easier to use with rear defection resulting in forward motion.
Practicality	5	2	2	Chin joystick and sip and puff control needs to be removed for transfers, feeding. Chin joystick needs to be mounted so that it is in the optimum position for the user; if the joystick is moved from this position then the control system is inoperable. Sip and puff has only limited commands available and is not proportional control.
Efficiency	5	2	3	Sip and puff has only limited commands available and is not proportional. Head and neck movements are more difficult than hand movements to control with precision. Chin control is proportional but the neck muscles fatigue more easily than the hand muscles.
Reliability	5	3	3	Sip and puff and chin joystick mounting bracketry are prone to breakages and wear. Sip and puff straws can become clogged.
Ease of installation and set up	5	3	3	Hand joysticks need to be adjusted so they are in the right position for the user but are generally very simple to set up. Chin control and sip and puff are time consuming to set up and require a trained clinician/technician.
Ease of daily set up	5	4	3	The sip and puff straw and the chin joystick need to be mounted so that they are in the optimum position for the user; if the straw or joystick is moved from this position then the control systems are inoperable.
Inexpensive	5	3	4	Due to economies of scale the hand joystick is relatively inexpensive. Sip and puff and chin joysticks are relatively more expensive.
Total	39	20	24	

ever, based on the limited availability of other control types on the manufacturers' standard prescription forms and the authors' own experiences, this is probably underestimated. None of the four studies examined are completely representative of the whole population of EPW users. Three cover only one particular patient group; either those requiring specialist seating (who are likely to requiring more specialised control devices) or those with spinal cord injuries (who, depending on level of injury, may not be able to operate a hand joystick). The other study draws on an apparently larger and more representative group of patients, but the methodology employed means that quality of the data provided to the authors is open to question.

One of the limitations of the review of the prescription forms for EPWs and the control devices available was that it was limited to EPWs available to the NHS contract in Scotland, which means that it may not be a fair representation of what EPWs are available to other countries. The companies that provide the prescription are, however, multinational and therefore it is likely the same EPWs with similar control device options will be available around the world. The Scottish contract does not include outdoor-only EPWs and this is another limitation of the availability review. Still, it should be noted that outdoor-only EPWs, as opposed to mobility scooters, are a small minority of EPWs and are generally specialised devices for a niche market.

It is thought that hand control joysticks will remain the most commonly used type of control device; however, patients with insufficient hand muscle strength and dexterity, hand tremors and upper limb spasticity are unable to use joysticks or even more specialised joysticks (e.g. isometric). One of the two most frequently used alternative control devices is the sip and puff control device and this offers patients who have limited head and neck movement or strength a control method for an EPW. However, when compared to standard hand control joysticks it had several drawbacks, particularly in terms of safety, intuitiveness, practicality and efficiency. The other main alternative control device is the chin joystick, this offers proportional control and therefore has comparable efficiency in movement to a traditional hand joystick. On the other hand, when compared to standard hand control joysticks it had several drawbacks, particularly in terms of safety and practicality. Due to these drawbacks there will a percentage of patients who are unable or who do not wish to use either of these two control methods. Other control methods are available but this study demonstrates they are infrequently used.

The terminology used by the studies examined was found not to be consistent. It was, therefore, occasionally necessary to interpret and make assumptions about the specific nature of the devices reported. The authors are fairly confident that their interpretations were correct, but there remains a potential risk that the some of the devices may have been wrongly categorised impacting on the validity of the results. This highlights the importance of a common, accepted vocabulary to support and foster communication amongst individuals working in all aspects of the wheelchair field and wheelchair users, their families and support workers [4]. In the area of control devices for EPW, the range and type of devices is growing rapidly and there is particular need for a well defined vocabulary. This would help clinicians and others better understand the potential of newer, novel control devices. The definitions developed and presented in Table 1 will, the authors hope, help in this regard.

In light of the drawbacks of current control devices, it must be expected that there are a proportion of patients who will have tried the control devices discussed in this paper and will have subsequently abandoned using them. None of the studies reviewed quantified this group of patients. One reason for this potential abandonment is the lack of adequate training in using the control methods. A case report by Bailey and coworkers [34] demonstrates that an extended period of training time is required to be able to teach a patient with physical and cognitive impairments the new skills to enable them to drive an EPW. If this time is not available or if there is not the motivation on behalf of the clinician or the patient then abandonment will be the probable outcome. An alternative novel device might be indicated if it reduces training time and therefore reduces the probability of abandonment. Another reason for abandonment is that control devices are just not suitable for the patient's needs and this is another indication for a novel device.

Another proportion of patients will have been assessed for an EPW but will have not received one because they do not meet the qualifying or eligibility criteria. This could be due to insufficient motor skills, strength, visual acuity, or due to fact they can operate a manual wheelchair indoors. Training amongst clinicians is an important factor that could influence whether a patient is considered for an EPW and the type of control devices being issued. If clinicians do not consider or have no knowledge of alternative non-hand joystick controls before potentially assessing a patient for an EPW then they may take the decision that the patient will fail the qualifying criteria and then subsequently not undertake the assessment for the EPW. It is important that clinicians who are able to refer a patient for an EPW assessment are familiar with all different types of commonly available control devices, their advantages, disadvantages, indications, and contra-indications.

5. Conclusion

The standard, hand joystick meets the control device needs of the vast majority of current EPW users. They are intuitive and easy to use, have excellent proprioceptive feedback and safe as they automatically return to their neutral position when released. They are relatively simple to manufacture in high volume and so are also very cost effective. There are, however, contra-indications for the use of a standard, hand joystick including poor hand muscle strength, proprioception, mobility and dexterity, hand tremors, and spasticity in the upper limbs. The most popular alternative control devices are sip and puff and chin joysticks, yet when they were compared to the standard, hand joystick using the literature search results and the authors' own experiences they performed poorly and both had several practical and aesthetic drawbacks.

The terminology used to describe control devices in journal articles and prescription forms was found to be inconsistent. This is counter productive for increasing awareness amongst clinicians of the control devices available. Clinicians need to be aware of all the different control devices available and their advantages, disadvantages, indications and contra-indications; this will enable them to prescribe a control device that will, as far as possible, meet the patient's individual requirements by informing their decision making, and ultimately reduce the chance of device abandonment.

There is need for future work in this area. The prevalence data is limited and possibly skewed by non-representative groups of patients. There is a lack of information on those who are, for whatever reasons, unable to use an EPW. There are numerous, emerging control devices and these need to be critically evaluated to determine which are most likely to address the drawbacks of the current control devices and will meet the unmet needs of potential EPW users.

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